

B.C.A. (Sem. II) Examination, 2026

(Session : 2025–28)

2 BCA 1

Digital Computer Organization

Paper Code : 211601

Time : 3:00 Hours

[Full Marks : 100]

[Pass Marks : 40]

Note : Candidates are required to give their answers in their own words as far as practicable. The questions are of equal value. Answer any **five** questions.

1. Convert the following numbers into equivalent binary numbers:

(a) $(45.25)_{10} = (\quad)_2$

(b) $(73)_8 = (\quad)_2$

(c) $(156)_{10} = (\quad)_2$

(d) $(111010.101)_2 = (\quad)_8$

2. Perform the following additions using 8-bit 2's complement arithmetic. Also state whether overflow occurs:

(a) $+38$ and $+47$

(b) -28 and $+16$

(c) $+72$ and $+61$

(d) -55 and -30

3. What is a logic gate? Explain AND, OR and NOT gates with their truth tables. Also describe the concept of a universal gate.

4. Define Boolean Algebra. State any five basic laws of Boolean algebra and verify any two laws with suitable examples.

5. Explain any **two** of the following:

(a) Principle of duality

(b) De Morgan's laws

(c) Karnaugh Map

(d) BCD

6. Draw and explain a Half-Adder circuit. Show how a Half-Adder can be constructed using basic logic gates and give its truth table.

7. (a) Simplify the following Boolean expressions:

(i) $A(B + C') + A'BC$

(ii) $(A + B)(A' + C)(B + C)$

(b) Find the complement of the following expressions:

(i) $(A + B')C + D'$

(ii) $(X' + Y + Z)(X + Y')(Y + Z')$

8. Explain Karnaugh Map (K-Map) in detail. Describe the procedure for simplifying Boolean expressions using a K-Map. Simplify the following Boolean function using a 4-variable K-Map and draw the corresponding logic circuit:

$$F(A, B, C, D) = \sum(0, 2, 3, 6, 7, 8, 10, 11, 14, 15)$$

9. What is a Flip-Flop? Explain the S-R Flip-Flop with a suitable logic diagram and Truth table. Mention its applications.
10. Write short notes on any **two** of the following:
- (i) Registers
 - (ii) Multiplexers
 - (iii) Sequential circuits
 - (iv) ROM and RAM

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2 BCA 2

Advanced Programming in C

Paper Code : 211602

Time : Three Hours

[Full Marks : 100]

[Pass Marks : 40]

Note : Candidates are required to give their answers in their own words as far as practicable. The questions are of equal value. Answer any **five** questions.

1. Differentiate structure and union with suitable example. Also explain Array in structure using program.
2. Write a C program using an array of structures to store details of 10 books such as Book-ID, Title, Author and Price. Display the details of books whose price is greater than a value entered by the user.
3. Explain dynamic memory allocation in C. Describe `malloc()`, `calloc()`, `realloc()` and `free()` with syntax and examples.
4. Explain any **two** of the following with suitable examples:
 - (a) Enumerations and typedef
 - (b) Nested structures
 - (c) Self-referential structures
 - (d) Chain pointers
5. How Pointers are created and initialized? Explain Advantages of pointer and call by value and call by reference.
6. Explain recursion in C. Write recursive functions to generate the first n terms of the Fibonacci sequence.
7. Explain the following concepts related to function and pointers in C, with suitable c examples:
 - (i) Function returning a pointer
 - (ii) Passing an array to a function
 - (iii) Pointer to a function
 - (iv) Passing Array of Structure
8. Compare any **two** of the following pairs in detail, giving suitable examples:
 - (i) Text file and Binary file
 - (ii) `fprintf()/fscanf()` and `fwrite()/fread()`
 - (iii) Static memory allocation and Dynamic memory allocation
9. Explain any five string related functions. Write a program for checking palindrome string.
10. What is recursion? Write a program to find factorial of a number using recursion.

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2 BCA 3

Fundamental Data Structure

Paper Code : 211603

Time : Three Hours

[Full Marks : 100]

[Pass Marks : 40]

Note : Candidates are required to give their answers in their own words as far as practicable. The questions are of equal value. Answer any **five** questions.

1. Explain the need of data structure. Explain binary search with the help of program.
2. Explain data structure based on FIFO with program and real life examples.
3. Explain the concepts of binary tree. Explain traversing algorithm of binary tree.
4. What is reverse Polish Notation? Convert the following infix notation into post-fix and pre-fix notation :

(a) $A + [B / (C + (D - E))] \% F / G - H$

Give stepwise representation.

5. Write a program for insertion and deletion in queue.
6. Write as program for creation of node and traversing a single linked list.
7. Explain Dynamic memory allocation with suitable example.
8. How an element can be inserted at the middle in Single Link list.
9. How an element can be sorted in array using sequential sort?
10. Draw a binary tree with following sequence:

PRE ORDER : A B C D E F

POST ORDER : B H G A C D E F

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2 BCA 4

Application Programming in Fox Pro

Paper Code : 211604

Time : 3:00 Hours

[Full Marks : 100]

[Pass Marks : 40]

Note : Candidates are required to give their answers in their own words as far as practicable. The questions are of equal value. Answer any **five** questions.

1. Define a Database Management System and explain the major characteristics of a relational database. Compare the database approach with the traditional file-processing approach.
2. Write a FOXPRO program to display the multiplication table of a number entered by the user, from 1 to 10.
3. Describe the different data types supported by FoxPro. Give the purpose and typical use of each data type with suitable examples.
4. Explain the use of conditional and looping statements in FoxPro. Illustrate IF....ELSE, DO WHILE and FOR...ENDFOR with suitable examples.
5. Develop a FoxPro program that accepts 10 numbers into an array, identifies the largest and smallest values, and displays their difference.
6. Explain the following FoxPro concepts:
 - (i) User-defined Function
 - (ii) Parameter Passing
 - (iii) Index File
 - (iv) Record Sorting
7. Distinguish between a procedure, a function and a command in FoxPro. Write a user-defined function to determine whether a given number is prime or not.
8. Write the syntax and explain the purpose of the following FoxPro Interface commands:
 - (i) DEFINE WINDOW
 - (ii) DEFINE MENU
 - (iii) DEFINE POPUP
 - (iv) @...SAY/GET commands
9. Write short notes on any **two** of the following:
 - (i) Database File and Table
 - (ii) Fields, Records and Memo Fields
 - (iii) Index Expressions and Tags
 - (iv) Reports and Labels
10. Explain the Fox Pro commands used to display and manipulate graphical elements such as boxes and lines. Write suitable command examples for each.

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2 BCA 5

Financial Accounting

Paper Code : 211605

Time : 3:00 Hours

[Full Marks : 100]

[Pass Marks : 40]

Note : Candidates are required to give their answers in their own words as far as practicable. The questions are of equal value. Answer any **five** questions.

1. What is accounting? Explain its objectives and the role it plays in a business organisation.
2. What are source documents? Explain the importance of invoices, receipts and bills in accounting records.
3. Define ledger. Explain posting from journal to ledger and state the importance of ledger accounts.
4. Explain the following:
 - (a) Capital and Revenue Expenditure
 - (b) Assets and Liabilities
 - (c) Gross Profit and Net Profit
 - (d) Outstanding and Prepaid Expenses
5. What is a journal? Explain the rules of debit and credit and give suitable examples of journal entries.
6. What is a cash book? Describe single-column, double-column and triple-column cash books.
7. What are accounting concepts and conventions? Explain any five important concepts with examples.
8. What is a voucher? Explain the main types of accounting vouchers and their uses.
9. Prepare a double-column cash book from any 15 suitable transactions involving cash and bank, and show the closing balances.
10. What is a Trading Account? Explain its preparation and the calculation of gross profit or gross loss with an example.